

# **AI Prompt History Document**

**Project:** Peblo AI Backend Engineer Challenge – Mini Content Ingestion + Adaptive Quiz Engine

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## **1. Introduction**

This document records the prompts used with AI-assisted tools during the development of the assignment for the backend engineering challenge provided by Peblo.

AI tools were used primarily as a support resource for research, understanding backend architecture, and structuring documentation. The final system design and project explanation were organized manually based on the assignment requirements.

The purpose of this document is to provide transparency regarding how AI assistance was used during the development process.

## **2. AI Tools Usage**

During the development of the assignment, AI-assisted language models were used to help with:

- Understanding backend system architecture
- Structuring documentation sections
- Generating example data formats
- Reviewing API design approaches
- Understanding adaptive learning mechanisms

The AI outputs were reviewed, edited, and structured before being included in the final assignment report.

### 3. Prompt History

Below are examples of prompts used during the development of the assignment.

#### Prompt 1

Explain how a backend system can ingest educational PDF content and convert it into structured data for further processing.

##### Purpose

To understand how a content ingestion pipeline works and how PDF documents can be converted into structured text segments.

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#### Prompt 2

Provide an example JSON structure for storing extracted text chunks from educational documents.

##### Purpose

To design a structured format for storing extracted educational content before generating quiz questions.

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#### Prompt 3

Generate an example JSON format for a multiple choice quiz question generated from educational content.

##### Purpose

To understand how generated quiz questions should be structured before storing them in the database.

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#### Prompt 4

Explain how adaptive quiz difficulty works in educational learning systems.

## **Purpose**

To understand the concept of adaptive learning and how quiz difficulty levels can be adjusted based on student performance.

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## **Prompt 5**

Provide a simple REST API structure for a quiz system that supports content ingestion, quiz generation, and answer submission.

## **Purpose**

To understand how API endpoints should be designed for interacting with the backend system.

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## **4. How AI Assistance Was Used**

AI-assisted tools were used mainly for:

- Understanding backend workflow design
- Structuring documentation sections
- Reviewing example API endpoint formats
- Generating example JSON structures
- Learning about adaptive quiz systems

All outputs generated by AI were reviewed and adapted to match the assignment requirements.

## **5. Responsible Use of AI**

AI tools were used as a support tool for learning and documentation assistance. The final project structure, explanations, and formatting were manually reviewed and organized to ensure accuracy and clarity. The goal was to use AI to assist in understanding system design concepts while maintaining originality in the assignment submission.

## **6. Conclusion**

The use of AI-assisted tools helped improve the efficiency of research and documentation during the assignment. By combining AI support with manual review and organization, the final submission demonstrates an understanding of backend architecture, AI integration, and API-based system design.

This prompt history provides transparency regarding how AI assistance contributed to the development of the assignment.